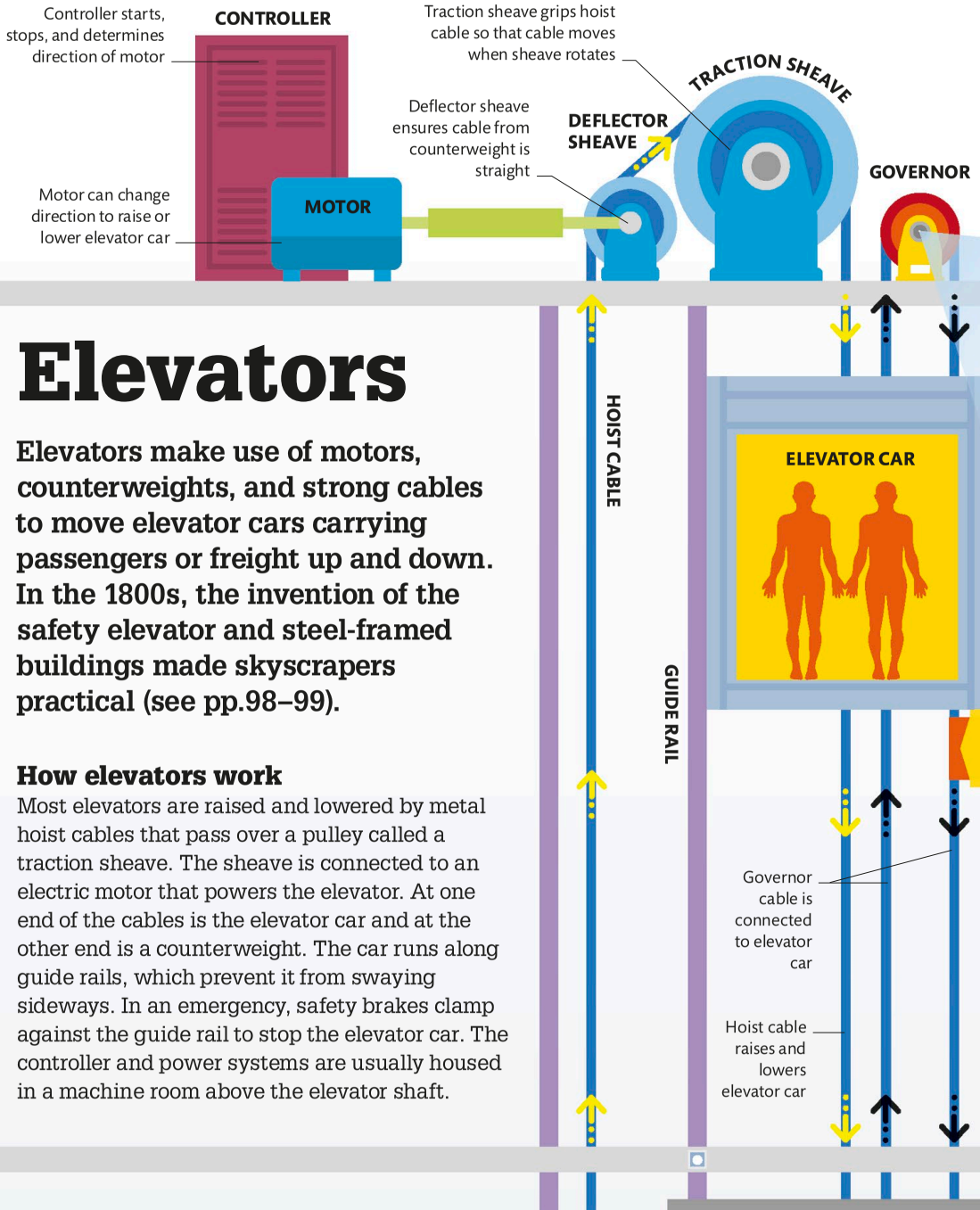


Elevators

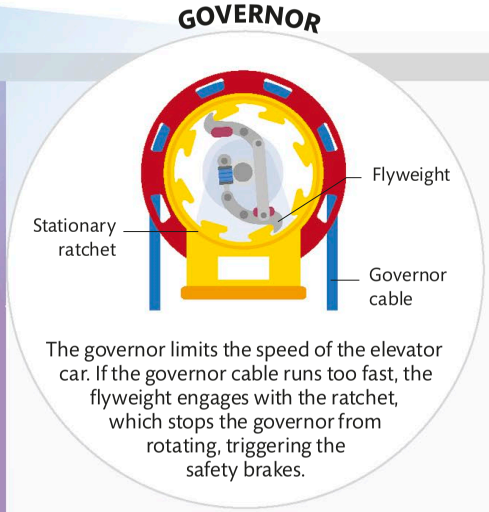
Elevators make use of motors, counterweights, and strong cables to move elevator cars carrying passengers or freight up and down. In the 1800s, the invention of the safety elevator and steel-framed buildings made skyscrapers practical (see pp.98–99).

How elevators work

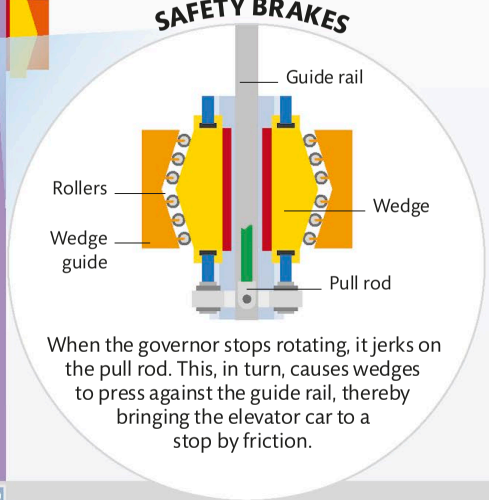
Most elevators are raised and lowered by metal hoist cables that pass over a pulley called a traction sheave. The sheave is connected to an electric motor that powers the elevator. At one end of the cables is the elevator car and at the other end is a counterweight. The car runs along guide rails, which prevent it from swaying sideways. In an emergency, safety brakes clamp against the guide rail to stop the elevator car. The controller and power systems are usually housed in a machine room above the elevator shaft.

Safety systems

All elevators have safety features that mean it is almost impossible for a car to plummet down a shaft. These include multiple cables, each of which can support the weight of a loaded car on its own, as well as speed control and safety brakes.



The governor limits the speed of the elevator car. If the governor cable runs too fast, the flyweight engages with the ratchet, which stops the governor from rotating, triggering the safety brakes.



When the governor stops rotating, it jerks on the pull rod. This, in turn, causes wedges to press against the guide rail, thereby bringing the elevator car to a stop by friction.